Extra material to the Course

Statistical Data Analysis with R

One Sample Tests

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Basing on the theory script of former statistic courses by Joachim Saborowski,

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One Sample t-Test

This document provides supplemental, in-dept information to the file `Statistical_Efficiency.pdf`. See also the attached code `t.test.step.by.step.R` and `normal.vs.t.distribution.R`

The One Sample t-Test is the model of choice when a random variable X with the observations $x_1 \dots x_n$ is to be compared with a known reference value μ_0 (e.g. given from literature or from experiments). The assumption is that $x_1, \dots, x_n$ are observations from an independent random variable with the properties

$$x_1 \dots x_n i.i.d. \sim N(\mu, \sigma^2),$$

which means that the random variable is approximately normally distributed with mean μ and variance σ^2. Note that the assumption refers to the random variable and not the actually surveyed observation. But since the true distribution of the random variable is not known (see `20231110_basic_statistical_vocabulary.pdf`), the distribution of the random sample is typically assumed to reflect the distribution of the random variable. In case the random sample contains more than 30 observations, normal distribution can be assumed generally. The distribution of the random samples with more than 30 observations thus needs not to be tested.

When the hypothesis of similarity of the random variable to the reference holds, the difference between μ and μ_0, a should be close to 0. This leads to the statistically testable hypotheses

$$H_0: a = 0 \quad \text{and} \quad H_1: a \neq 0.$$

Given the approximate normal distribution of the random variable, a suitable test statistic can be expressed as

$$t = \sqrt{n}\frac{\mu - \pi_0}{\sigma}.$$

Because the t-distribution approximates the normal distribution and also takes the sample size into account, in the form of the degrees of freedom. In practice, μ and σ^2 are not known. μ is estimated by the arithmetic mean and σ^2 by

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

The H_0 is rejected if the absolute test statistic value of (t-value) is greater than the $\left(1 - \frac{\alpha}{2}\right)$-quantile of the *t*-distribution with n-1 degrees of freedom ($|t| > t_{(n-1;1-\alpha/2)}$).

Wilcoxon Signed-Rank Test

See also the attached code `wilcoxon.test.step.by.step.R`

The Wilcoxon Signed-Rank Test is the non-parametric pendant to the t-Test. The assumptions of the Wilcoxon Signed-Rank Test are still that the elements $x_1 \dots x_n$ are independent observations from a random variable X with a continuous symmetric distribution. But in difference to the t-Test, the distribution can be arbitrary. The hypotheses can again be expressed as

$$H_0: a = 0 \quad \text{and} \quad H_1: a \neq 0,$$

since the observed values are again subtracted by a reference μ_0. The respective differences a surround 0 when the observations symmetrically deviate around μ_0.

The absolute values of the observations are ordered increasingly. The smallest value is defined as rank one, the greatest value as the rank *n*. The test static S is calculated as

$$S = \sum_{i=1}^{n^+} R_i^+ - \frac{n(n+1)}{4}$$

Whereas *n* is the number of observations and n^+ is the number of positive observations. R_i^+ are the ranks of the positive values for the a. The Wilcoxon Test is less efficient (less powerful) than the t-Test. This is a typical issue of non-parametric tests. It follows that the advantage of arbitrary on the one hand leads to a lack of efficiency on the other hand. The Wilcox Signed-Rank Test fails more often to reject the H_0 although it is truly rejected (type I error). In practice, this means that more observations are required to achieve meaningful results. This property is explained in a simple example in the code `Statistical_Efficiency.pdf`. It is shown that arithmetic mean (as a simple example for a parametric statistic) converges quicker to the known exact average than the median. The arithmetic mean therefore requires less observations to reflect a suitable outcome. Whenever possible, a parametric statistic is thus preferred.